Name
Date
Course

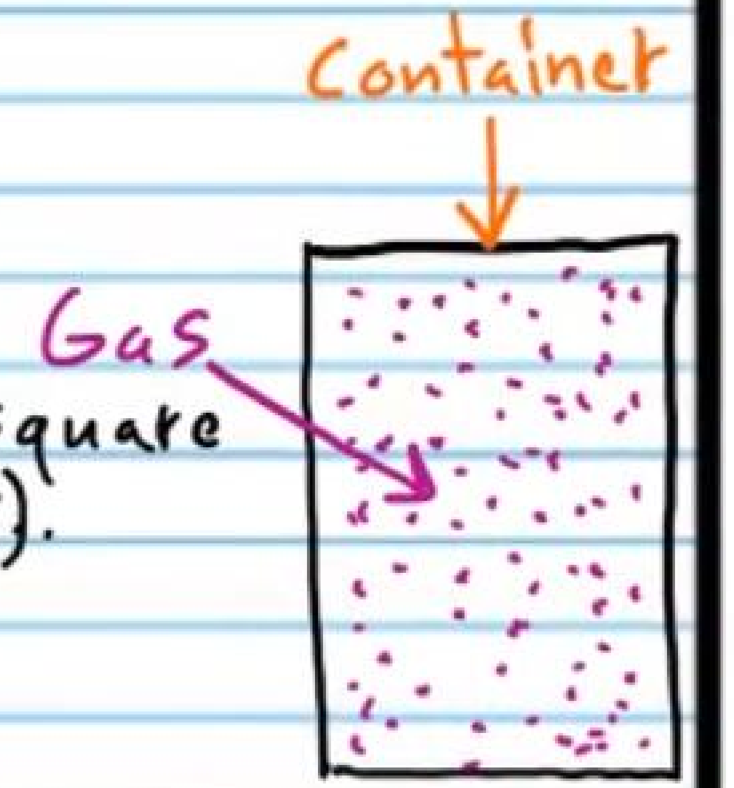
Applications of Algebra (Part I)

Direct and Inverse Proportionality

Direct Proportionality: If $y = kx$, where k is some constant, then we say that y is directly proportional to x. k is called the proportionality constant.

Inverse Proportionality: If $y = \frac{k}{x}$, where k is some constant, then we say that y is inversely proportional to x. k is called the proportionality constant.

① The pressure of the gas in the container at the right is directly proportional to the temperature of the gas, such that $P = 0.175T$, where P is pressure (in pounds per square inch) and T is temperature (in Kelvins).



a) If the temperature in the container is 312 Kelvins, what is the pressure (in pounds per square inch)?

$y = kx$

$P = 0.175T$

$P = 0.175(312)$

$\boxed{P = 54.6 \text{ psi}}$

b) What is the proportionality constant?

$\boxed{0.175}$

c) If the pressure in the container is 49 psi, what is the temperature of the gas?

$P = 0.175T$

$49 = 0.175T$

$\boxed{T = 280 \text{ Kelvins}}$

② The electrical current in a wire is directly proportional to the voltage provided, such that $I = 3500V$, where I is current (measured in amperes) and V is voltage (measured in volts).

a) If 8 volts are provided in the wire, how many amperes of electrical current are flowing?

$y = kx$

$I = 3500V$

$I = 3500(8)$

$\boxed{I = 28{,}000 \text{ amperes}}$

b) What is the proportionality constant?

$\boxed{3500}$

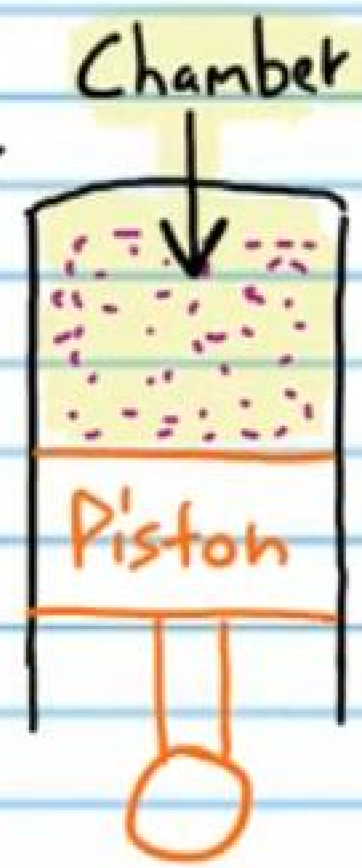
c) If there are 18,000 amperes of electrical current flowing in the wire, how many volts were provided?

$$I = 3500V$$

$$18{,}000 = 3500V$$

$$\boxed{V = 5.14 \text{ Volts}}$$

③ When the piston at the right moves up and down, it changes the volume of the cylinder chamber. The pressure inside the chamber is inversely proportional to the volume of the chamber, such that $P = 0.001/V$, where V is the volume of the chamber (in liters) and P is the gas pressure in the chamber (in pounds per square inch).



a) If the volume of the chamber is 0.43 liters, what is the pressure in the chamber?

$$y = \frac{k}{x}$$

$$P = \frac{0.001}{V}$$

$$P = \frac{0.001}{0.43} = \boxed{0.00233 \text{ psi}}$$

b) What is the proportionality constant?

$$\boxed{0.001}$$

c) If the pressure in the chamber is 0.05 psi, what is the volume of the cylinder chamber?

$$P = \frac{0.001}{V}$$

$$0.05 = \frac{0.001}{V}$$

$$(V)\,0.05 = \frac{0.001}{V}(V)$$

$$0.05V = 0.001$$

$$\boxed{V = 0.02 \text{ liters}}$$

④ Sometimes proportionality constants are not determined experimentally, but are simply known by context. For example, the distance between my house and the beach is 7 miles. The time that it takes for me to ride my bicycle from my house to the beach is inversely proportional to the speed that I ride, such that $t = 7/r$, where t is time (in hours) and r is speed in miles per hour. In this case, 7 is the constant, but it is not necessary to perform experiments to find it. It is simply the distance between my house and the beach.

a) If I ride 12 miles per hour, how long will it take me to get to the beach? Assume that my speed does not change during the entire trip.

$$y = \frac{K}{X}$$

$$t = \frac{7}{r}$$

$$\boxed{t = \frac{7}{12} \text{ of an hour}}$$

b) What is the proportionality constant?

$\boxed{7}$

c) If it takes me 0.4 hours to ride to the beach at a constant pace, what was my speed?

$t = \frac{7}{r}$

$0.4 = \frac{7}{r}$

$(r)0.4 = \frac{7}{r}(r)$

$0.4r = 7$

$\boxed{r = 17.5 \text{ mph}}$

Two-Part Proportionality Problems

5. If y is directly proportional to x, and y=6 when x=14, find y when x=10.

I) $y = kx$

$6 = k(14)$

$\frac{3}{7} = k$

II) $y = \frac{3}{7}x$

$y = \frac{3}{7}(10)$

$\boxed{y = \frac{30}{7}}$

6. The cost of a phone is inversely proportional to the number that are sold. If 600,000 phones are sold for \$437 each, what would the cost per phone be if 250,000 were sold?

I) $y = \frac{k}{x}$

$C = \frac{k}{N}$

$437 = \frac{k}{600{,}000}$

$k = 262{,}200{,}000$

II) $C = \frac{262{,}200{,}000}{N}$

$C = \frac{262{,}200{,}000}{250{,}000}$

$\boxed{C = \$1{,}048.80}$

7. If y is inversely proportional to x, and x=5.6 when y=20, find y when x is 3.

I) $y = \frac{k}{x}$

$20 = \frac{k}{5.6}$

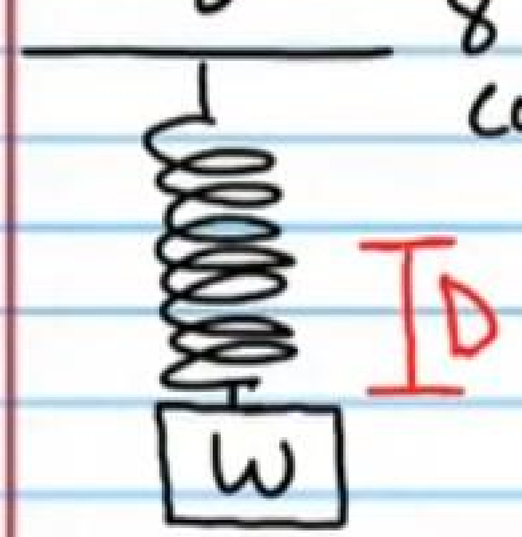
$k = 112$

II) $y = \frac{112}{x}$

$y = \frac{112}{3}$

$\boxed{y = 37.\overline{3}}$

⑧ An object is suspended from a ceiling with a spring. The distance D that the spring stretches is directly proportional to the weight W of the object. If the spring stretches 8 centimeters when a 1.8 pound object is connected to it, how much weight would be needed to stretch the spring 13 centimeters?



I) $y = KX$

$D = KW$

$8 = K(1.8)$

$K = 4.\overline{4}$

II) $D = 4.\overline{4}W$

$13 = 4.\overline{4}W$

$\boxed{W = 2.925 \text{ pounds}}$

Finding the Number of Coins with Systems of Three Equations

⑨ Jerry has only nickels, dimes, and quarters in his pocket. There are 34 coins worth a total of \$6.15. How many of each type of coin does he have if there are 3 more nickels than dimes?

Q = # of quarters

N = # of nickels

D = # of dimes

$$\begin{cases} a: & Q + N + D = 34 \\ b: & 0.25Q + 0.05N + 0.1D = 6.15 \\ c: & N = D + 3 \end{cases}$$

$$\begin{cases} a: & Q + D + 3 + D = 34 \\ b: & 0.25Q + 0.05(D+3) + 0.1D = 6.15 \end{cases}$$

$$\begin{cases} a: & Q + 2D + 3 = 34 \\ b: & 0.25Q + 0.05D + 0.15 + 0.1D = 6.15 \end{cases}$$

$$(0.25)\begin{cases} a: & Q + 2D = 31 \\ b: & 0.25Q + 0.15D = 6 \end{cases}$$

$$\begin{cases} a: & 0.25Q + 0.5D = 7.75 \\ b: & 0.25Q + 0.15D = 6 \end{cases}$$

$0.35D = 1.75$

$\boxed{D = 5 \text{ dimes}}$

$N = 5 + 3$

$\boxed{N = 8 \text{ nickels}}$

$Q + 2(5) = 31$

$Q + 10 = 31$

$\boxed{Q = 21 \text{ quarters}}$

⑩ Cynthia found 57 coins that are nickels, dimes, and quarters only. The number of dimes is one less than twice the number of quarters. If the total worth of the coins is \$6.40, how many of each type of coin are there?

Q = # of quarters

N = # of nickels

D = # of dimes

$$\begin{cases} a: & Q+N+D=57 \\ b: & 0.25Q+0.05N+0.1D=6.40 \\ c: & D=2Q-1 \end{cases}$$

$$\begin{cases} a: & Q+N+2Q-1=57 \\ b: & 0.25Q+0.05N+0.1(2Q-1)=6.40 \end{cases}$$

$$\begin{cases} a: & 3Q+N-1=57 \\ b: & 0.25Q+0.05N+0.2Q-0.1=6.40 \end{cases}$$

$$\begin{cases} (0.05)\ a: & 3Q+N=58 \\ b: & 0.45Q+0.05N=6.50 \end{cases}$$

$$\begin{cases} a: & 0.15Q+0.05N=2.9 \\ b: & 0.45Q+0.05N=6.50 \end{cases}$$

$$-0.3Q=-3.6$$

$$\boxed{Q=12 \text{ quarters}}$$

$$D=2(12)-1$$

$$D=24-1$$

$$\boxed{D=23 \text{ dimes}}$$

$$12+N+23=57$$

$$N+35=57$$

$$\boxed{N=22 \text{ nickels}}$$

⑪ A cash register has only quarters, nickels, and dimes (no bills or pennies), worth a total of \$3.60. There are three times as many quarters as nickels. If there are 24 coins total, how many of each type of coin are in the register?

Q = # of quarters

N = # of nickels

D = # of dimes

$$\begin{cases} a: & Q+N+D=24 \\ b: & 0.25Q+0.05N+0.1D=3.6 \\ c: & Q=3N \end{cases}$$

$$\begin{cases} a: & 3N+N+D=24 \\ b: & 0.25(3N)+0.05N+0.1D=3.6 \end{cases}$$

$$\begin{cases} a: & 4N+D=24 \\ b: & 0.75N+0.05N+0.1D=3.6 \end{cases}$$

$$\begin{cases} a: & 4N+D=24 \\ b: & 0.8N+0.1D=3.6 \end{cases}$$

$$\begin{cases} a: & 4N+D=24 \\ b: & 4N+0.5D=18 \end{cases}$$

$$0.5D=6$$

$$\boxed{D=12 \text{ dimes}}$$

$$4N+12=24$$

$$\boxed{N=3 \text{ nickels}}$$

$$Q=3(3)$$

$$\boxed{Q=9 \text{ quarters}}$$

(12) A child collects quarters, dimes, and nickels lying in the pool of a Fountian. He leaves the pennies for someone else. He collects a total of 39 coins worth \$5.30. If the number of dimes is 4 more than the number of nickels, how many of each type of coin are there?

Q = # of quarters

N = # of nickels

D = # of dimes

$$\begin{cases} Q+N+D=39 & (a) \\ 0.25Q+0.05N+0.1D=5.30 & (b) \\ D=4+N & (c) \end{cases}$$

$$\begin{cases} Q+N+(4+N)=39 & (a) \\ 0.25Q+0.05N+0.1(4+N)=5.3 & (b) \end{cases}$$

$$\begin{cases} Q+2N+4=39 & (a) \\ 0.25Q+0.05N+0.4+0.1N=5.3 & (b) \end{cases}$$

$$(0.25)\ \begin{cases} Q+2N=35 & (a) \\ 0.25Q+0.15N=4.9 & (b) \end{cases}$$

$$\begin{cases} 0.25Q+0.5N=8.75 & (a) \\ 0.25Q+0.15N=4.9 & (b) \end{cases}$$

$$0.35N=3.85$$

$$\boxed{N=11 \text{ nickels}}$$

$$D=4+11$$

$$\boxed{D=15 \text{ dimes}}$$

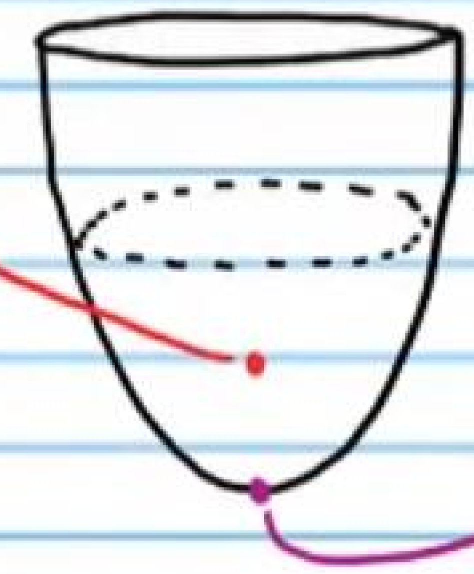
$$Q+2(11)=35$$

$$\boxed{Q=13 \text{ quarters}}$$

Paraboloids

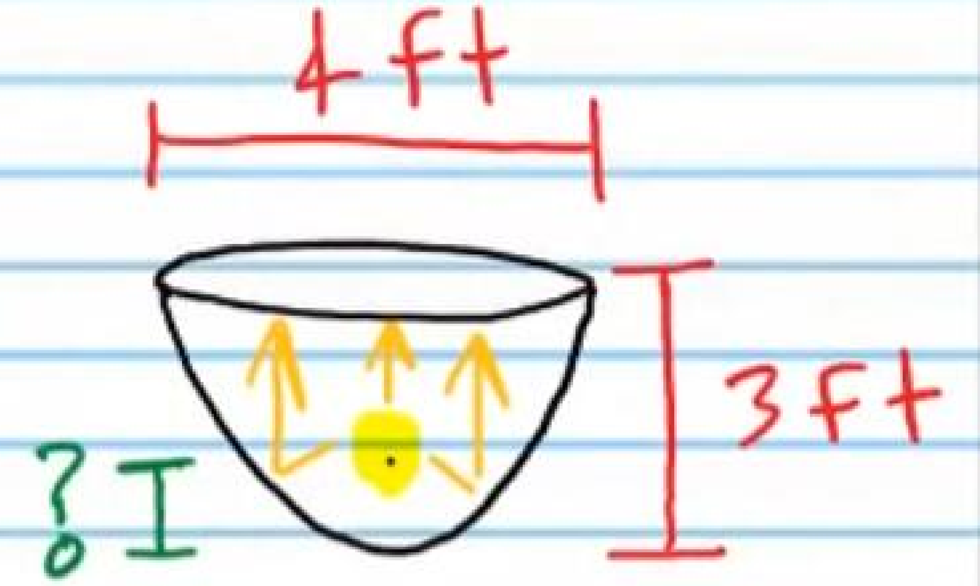
Paraboloid: A three-dimensional shape obtained by rotating a parabola about its line of symmetry, assuming that a thin solid is left in its trail.

The focus of a paraboloid is an important location for many applications.



Vertex of the paraboloid.

(13) A searchlight is shaped like a paraboloid. If the opening is 4 feet wide and the depth of the recess is 3 feet, where should the light be placed so that the rays are reflected in the right direction?



I) $y=a(x-h)^2+k$

$3=a(2-0)^2+0$

$3=a(2)^2+0$

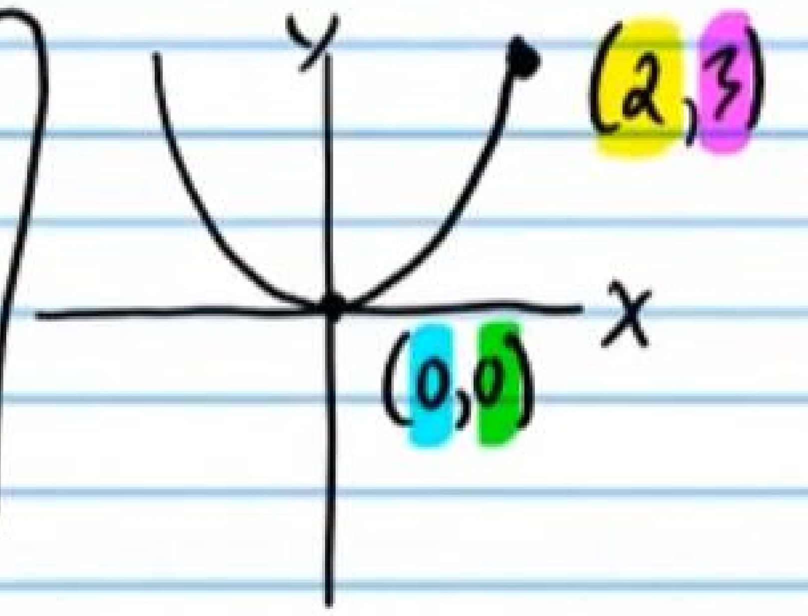
$3=4a$

$a=\frac{3}{4}$

II) $P=\frac{1}{4a}$

$P=\frac{1}{4\left(\frac{3}{4}\right)}$

$P=\frac{1}{3}$



The light should be placed $\frac{1}{3}$ of a foot from the vertex of the paraboloid.

(14) A satellite dish is shaped like a paraboloid. Signals from a satellite hit the dish and are reflected to a single point where a receiver is located. If the receiver is 2 meters from the vertex of the dish, and the opening of the dish is 6 meters wide, how long (i.e. deep) should the dish be?

I) $P = \frac{1}{4a}$

$2 = \frac{1}{4a}$

$(4a)2 = \frac{1}{4a}(4a)$

$8a = 1$

$a = \frac{1}{8}$

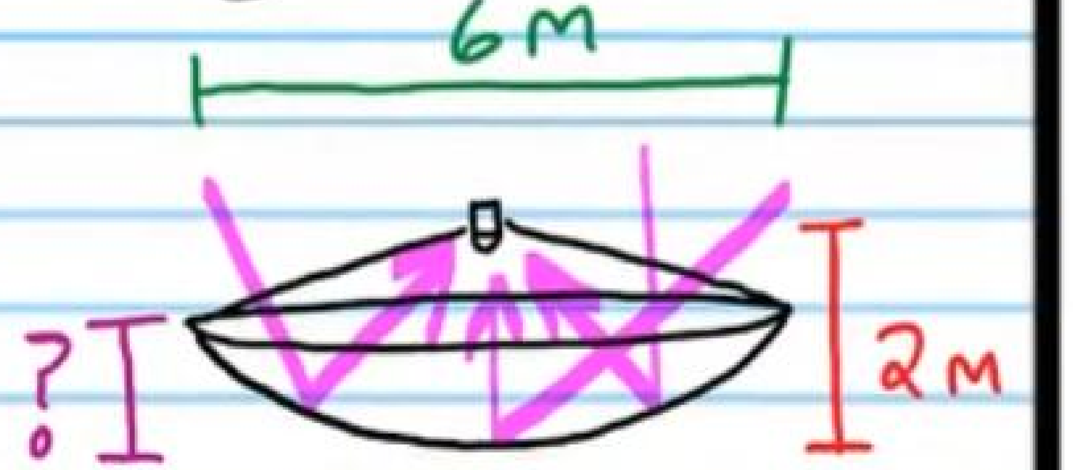
II) $y = a(x-h)^2 + k$

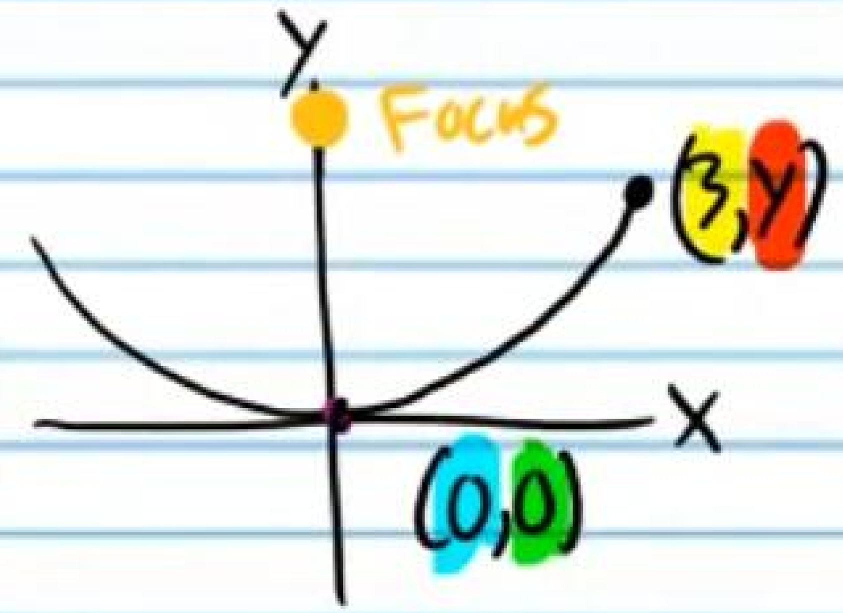
$y = \frac{1}{8}(3-0)^2 + 0$

$y = \frac{1}{8}(3)^2 + 0$

$y = \frac{9}{8} = 1\frac{1}{8}$

The dish should be $1\frac{1}{8}$ meters long.





(15) A searchlight is shaped like a paraboloid. If the paraboloid is 9 feet long and the light source is placed one foot from the vertex of the paraboloid, how wide should the opening of the paraboloid be?

I) $P = \frac{1}{4a}$

$1 = \frac{1}{4a}$

$(4a)1 = \frac{1}{4a}(4a)$

$4a = 1$

$a = \frac{1}{4}$

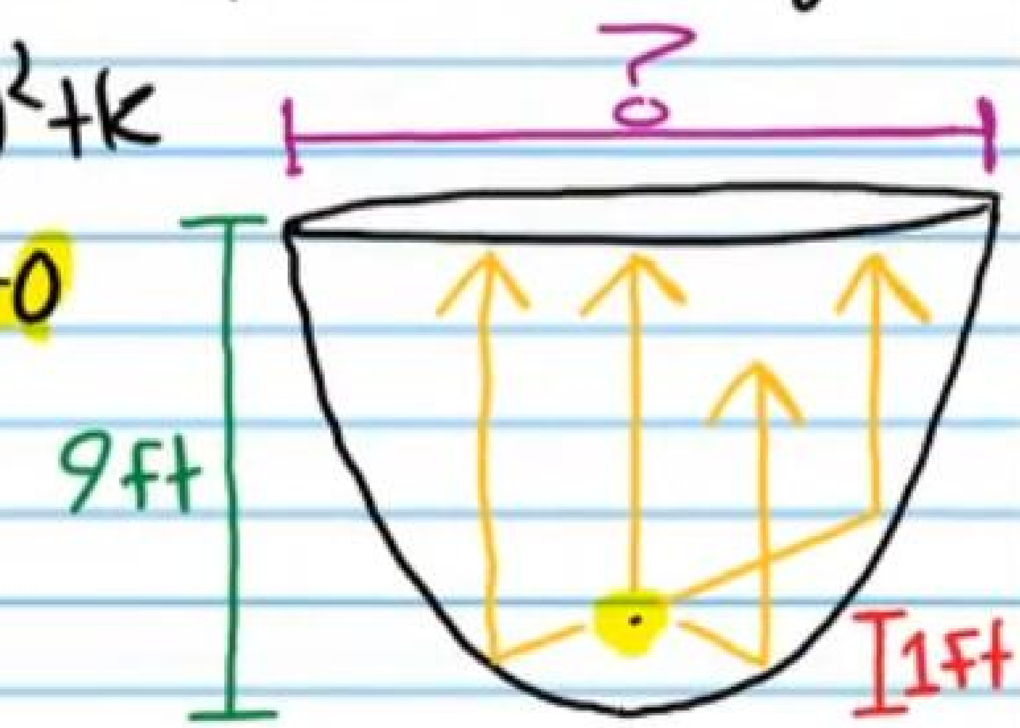
II) $y = a(x-h)^2 + k$

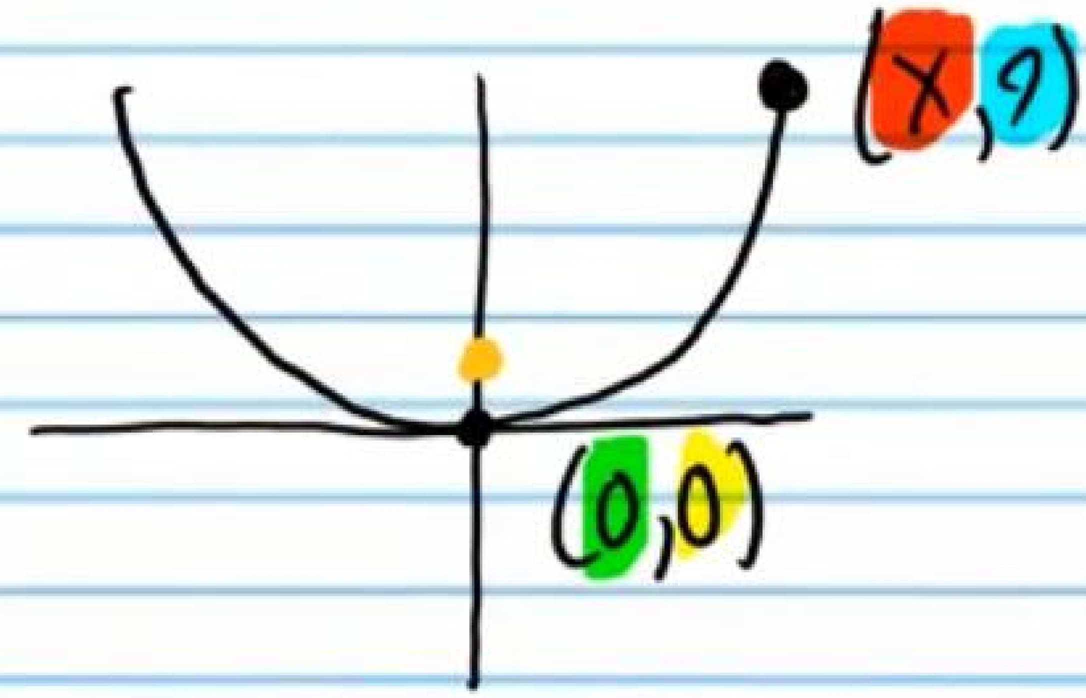
$9 = \frac{1}{4}(x-0)^2 + 0$

$9 = \frac{1}{4}x^2$

$36 = x^2$

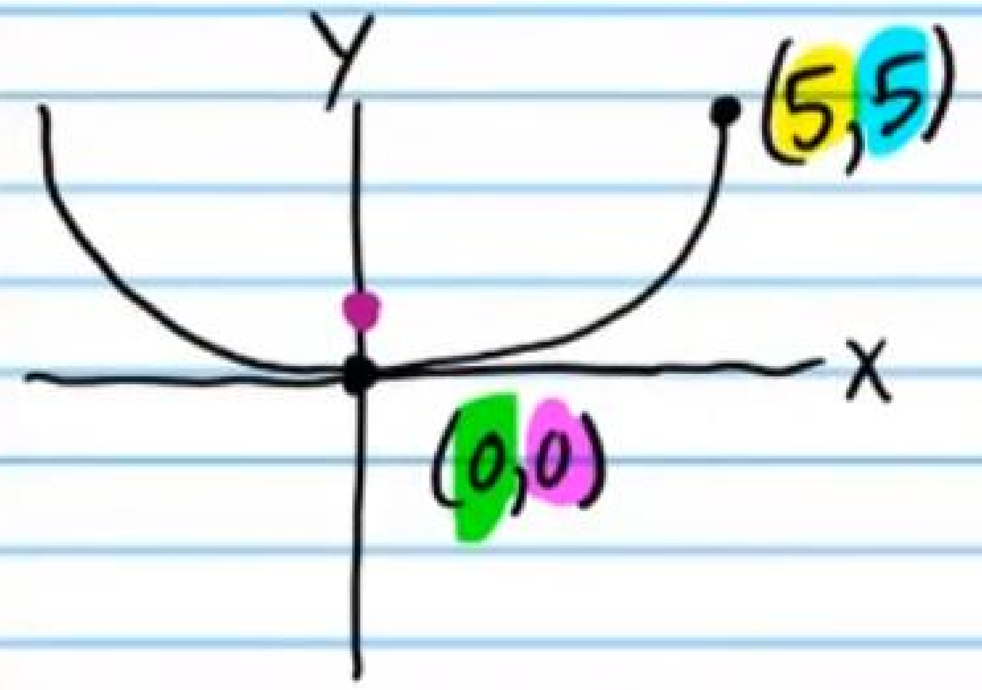
$x = 6$

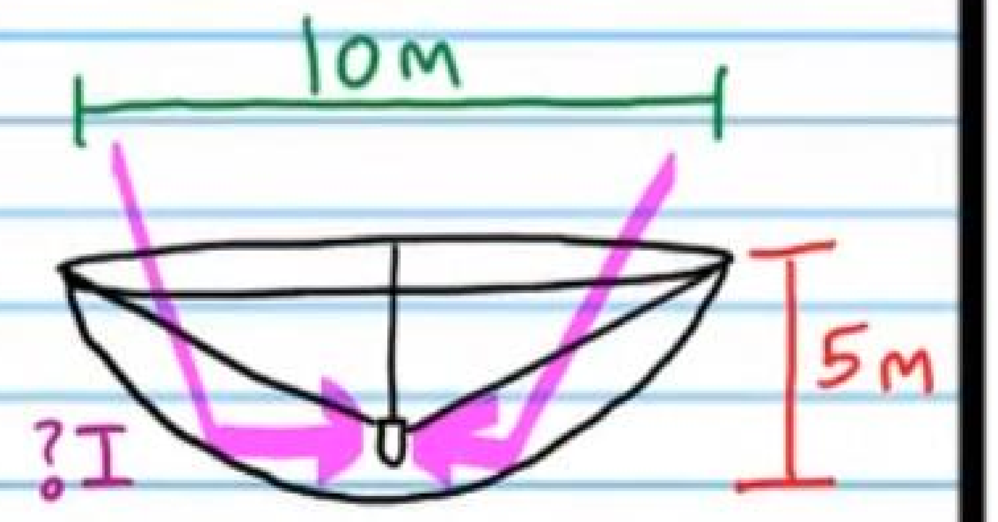




The opening of the paraboloid should be 12 feet wide.

(16) A satellite dish is shaped like a paraboloid. Signals from a satellite hit the dish and are reflected to a single point where a receiver is located. If the dish is 5 meters long, and the opening of the dish is 10 meters, where should the receiver be placed?





I) $y = a(x-h)^2 + k$

$5 = a(5-0)^2 + 0$

$5 = a(5)^2$

$5 = 25a$

$a = 1/5$

II) $P = \frac{1}{4a}$

$P = \frac{1}{4(1/5)}$

$P = \frac{1}{\frac{4}{5}}$

$P = \frac{5}{4}$

$\boxed{\text{The receiver should be placed } 1\tfrac{1}{4} \text{ meters from the vertex of the paraboloid.}}$